MTH 418a: Inference-I
Assignment No. 3: Minimal Sufficiency and Completeness

1. Show that every function of a complete statistic is complete.

2. Let $X_1, \ldots, X_n$ be a random sample from a population with p.d.f./p.m.f. $f_\theta(\cdot), \theta \in \Theta$. In each of the following cases, show that $T(\underline{X})$ is a minimal sufficient statistic. Also verify if it is complete.

   (a) For known $\mu_0 \in \mathbb{R}, X_1 \sim \mathrm{N}(\mu_0, \theta^2), \Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n (X_i - \mu_0)^2$. Is $\underline{U}(\underline{X}) = (\overline{X},\ \sum_{i=1}^n (X_i - \overline{X})^2)$ sufficient and complete?;
   (b) For known $\sigma_0 > 0$, $X_1 \sim \mathrm{N}(\theta, \sigma_0^2), \Theta = \mathbb{R}$, $T(\underline{X}) = \sum_{i=1}^n X_i$;
   (c) $X_1 \sim \mathrm{N}(\mu, \sigma^2), \underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty)$, $\underline{T}(\underline{X}) = (\overline{X},\ \sum_{i=1}^n (X_i - \overline{X})^2)$. Is $U(\underline{X}) = \overline{X} + S^2$ sufficient and complete?;
   (d) For known $\mu_0 \in \mathbb{R}, X_1 \sim \mathrm{Exp}(\mu_0, \theta), \Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n (X_i - \mu_0)$. Is $U(\underline{X}) = \overline{X}$ complete and sufficient?;
   (e) For known $\sigma_0 > 0$, $X_1 \sim \mathrm{Exp}(\theta, \sigma_0), \Theta = \mathbb{R}$, $T(\underline{X}) = X_{(1)}$. Is $\underline{U}(\underline{X}) = (X_{(1)},\ \sum_{i=1}^n (X_i - X_{(1)}))$ sufficient and complete?;
   (f) $X_1 \sim \mathrm{Exp}(\mu, \sigma), \underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty)$, $\underline{T}(\underline{X}) = (X_{(1)},\ \sum_{i=1}^n (X_i - X_{(1)}))$;
   (g) For known $\alpha_0 > 0, X_1 \sim \mathrm{Gamma}(\alpha_0, \theta), \Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n X_i$.

3. Let $X_1, \ldots, X_n$ be a random sample from a population with p.d.f./p.m.f. $f_\theta(\cdot), \theta \in \Theta$. In each of the following cases, show that $T(\underline{X})$ is a minimal sufficient statistic. Also verify if it is complete.

   (a) For known $\mu_0 \in \mathbb{R}, X_1 \sim \mathrm{N}(\mu_0, \theta^2), \Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n (X_i - \mu_0)^2$;
   (b) $X_1 \sim \mathrm{Beta}(\theta_1, \theta_2), \underline{\theta} = (\theta_1, \theta_2), \Theta = (0, \infty) \times (0, \infty)$, $T(\underline{X}) = (\prod_{i=1}^n X_i,\ \prod_{i=1}^n (1 - X_i)$);
   (c) $X_1 \sim \mathrm{N}(\theta, \theta), \Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n X_i^2$;
   (d) $X_1 \sim \mathrm{N}(\theta, \theta^2), \Theta = (0, \infty)$, $T(\underline{X}) = (\overline{X}, \sum_{i=1}^n (X_i - \overline{X})^2)$;
   (e) $X_1 \sim \mathrm{U}(\theta_1, \theta_2), \underline{\theta} = (\theta_1, \theta_2), \Theta = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 < x_2\}$, $T(\underline{X}) = (X_{(1)}, X_{(n)})$;
   (f) $X_1 \sim \mathrm{U}(\theta - \frac{1}{2}, \theta + \frac{1}{2}), \Theta = \mathbb{R}$, $T(\underline{X}) = (X_{(1)}, X_{(n)})$;
   (g) $f_\theta(x) = \frac{2(\theta - x)}{\theta^2} I_{(0,\theta)}(x), \Theta = (0, \infty)$, $T(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$;
   (h) $X_1 \sim \mathrm{Poisson}(\theta), \Theta = (0, \infty)$, $T(\underline{X}) = \overline{X}$;
   (i) For known positive integer m, $X_1 \sim \mathrm{Bin}(m, \theta), \Theta = (0, 1)$, $T(\underline{X}) = \sum_{i=1}^n X_i$;
   (j) For normalizing constant $c(\theta)$, $f_\theta(x) = \frac{c(\theta)}{x^2}$, $x = \theta+1,\ \theta+2, \ldots,$ $\Theta = \{0, 1, 2, \ldots\}$, $T(\underline{X}) = X_{(1)}$.

4. Each of the parts (a)-(j) below correspond to the corresponding parts of Problem 3. For each of the parts (a)-(j), determine whether the statistic $U(\underline{X})$, as defined below, is minimal sufficient for θ. Also verify if it is complete.

   (a) $U(\underline{X}) = (\overline{X},\ \sum_{i=1}^n (X_i - \overline{X})^2)$;
   (b) $U(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$;
   (c) $U(\underline{X}) = 15T^2(\underline{X}) + 8T(\underline{X}) - 23$;
   (d) $U(\underline{X}) = (\overline{X},\ \sum_{i=1}^n (X_i - 1)^2)$;

(e) $U(\underline{X}) = (2X_{(n)} - X_{(1)},\ X_{(n)} + 3X_{(1)})$;
(f) $U(\underline{X}) = (X_{(1)}, X_{(2)}, \ldots, X_{(n)})$;
(g) $U(\underline{X}) = (\prod_{i=1}^{n} X_i, X_{(2)} + X_{(3)},\ 2X_{2)} + 3X_{(3)},\ X_{(4)}, \ldots, X_{(n)})$;
(h) $U(\underline{X}) = 2\overline{X}^2 + 3\overline{X} - 100$;
(i) $U(\underline{X}) = \overline{X}^5$;
(j) $U(\underline{X}) = 9e^{2X_{(1)}} - 17e^{X_{(1)}} - 1083$.

5. Let X_1, X_2 be a random sample from a discrete uniform distribution on the set $\{\theta, \theta+1, \theta+2\}$, where $\theta \in \Theta = \{0, 1, 2, \ldots\}$. Show that $\underline{T} = (\frac{X_{(1)}+X_{(2)}}{2}, X_{(2)} - X_{(1)})$ is a minimal sufficient statistic. Is it complete?

6. Let X be a r.v. with distributional support $\chi = \{-1, 0, 1, 2, \ldots\}$ and p.m.f. $f_\theta, \theta \in (0,1) = \Theta$, where $f_\theta(-1) = \theta$ and $f_\theta(x) = (1-\theta)^2\theta^x,\ x = 0, 1, 2, \ldots$. Show that the family $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ is boundedly complete but not complete.

7. Let $X_1, X_2, \ldots, X_n$ be a random sample from a discrete uniform distribution on the set $\{1, 2, \ldots, \theta\}$, where $\theta \in \{1, 2, \ldots\} = \Theta$.

   (a) Show that that the family $\mathcal{P} = \{f_\theta : \theta \in \Theta\}$ is complete;
   (b) Let $m_0 \in \Theta$. Show that the family $\mathcal{P}_1 = \{f_\theta : \theta \in \Theta - \{m_0\}\}$ is not complete;
   (c) Show that $T(\underline{X}) = X_{(n)}$ is complete.

Honors Problems

8. Let $X_1, \ldots, X_n$ $(n \geq 2)$ be a random sample from $\text{Exp}(\mu, \sigma)$ distribution, where $\mu \in \mathbb{R}, \sigma > 0$. Show that $T_1(\underline{X}) = X_{(1)} \sim \text{Exp}(\mu, \frac{\sigma}{n})$ and $T_2(\underline{X}) = \sum_{i=1}^{n}(X_i - X_{(1)}) \sim \text{Gamma}(n-1, \sigma)$. Also, show that $T_1(\underline{X})$ and $T_2(\underline{X})$ are independently distributed. (**Hint:** Note that $T_2(\underline{X}) = \sum_{i=2}^{n}(X_{(i)} - X_{(1)})$)

9. Let $\mathcal{P}$ be a family of densities with common support, and let $\mathcal{P}_0 \subseteq \mathcal{P}$. If a statistic T is minimal sufficient for $\mathcal{P}_0$ and sufficient for $\mathcal{P}$, show that T is minimal sufficient for $\mathcal{P}$.

10. (a) Let $\mathcal{P} = \{f_0, f_1, f_2, \ldots, f_k\}$ be a family of densities, each having the same support. Prove that the statistic $\underline{T}(X) = (\frac{f_1(X)}{f_0(X)}, \frac{f_2(X)}{f_0(X)}, \ldots, \frac{f_k(X)}{f_0(X)})$ is minimal sufficient;
    (b) Let $\mathcal{P} = \{f_0, f_1, f_2\}$, where $f_0(x) = I_{(-1,0)}(x)$, $f_1(x) = I_{(0,1)}(x)$ and $f_2(x) = 2xI_{(0,1)}(x)$. Show that $\underline{T}(X) = (\frac{f_1(X)}{f_0(X)}, \frac{f_2(X)}{f_0(X)})$ is not minimal sufficient. Is it sufficient? Find a minimal sufficient statistic.

11. (a) Let X be a random sample from a population having p.m.f. $f_\theta,\ \theta \in (0,1) = \Theta$, where $f_\theta(x) = \frac{1}{4}$, if $x = 1, 2$; $= \frac{1+\theta}{4}$, if $x = 3$; $= \frac{1-\theta}{4}$, if $x = 4$; $= 0$, otherwise; $\theta \in \Theta$. Find a minimal sufficient statistic for θ. Is it complete?
    (b) Let X be a discrete r.v. with distributional support $\chi = \{-1, 0, 1\}$ and p.m.f. $f_\theta,\ \theta \in \Theta = \{1, 2, 3\}$. Suppose that $f_1(-1) = 0.4, f_1(0) = 0.2, f_1(1) = 0.4, f_2(-1) = 0.6, f_2(0) = 0.3, f_2(1) = 0.1, f_3(-1) = 0.2, f_3(0) = 0.1, f_3(1) = 0.7$. Find a minimal sufficient statistic for θ. Is it complete?

12. Let $X_1, \ldots, X_n$ be a random sample from a population with p.d.f. $f_\theta(\cdot), \theta \in \Theta$. In each of the following cases, find a minimal sufficient statistic $T(\underline{X})$. Verify if $T(\underline{X})$ is complete.

    (a) $f_\theta(x) = \frac{1}{\pi} \cdot \frac{1}{1+(x-\theta)^2},\ -\infty < x < \infty,\ \theta \in \mathbb{R} = \Theta;$
    (b) $f_\theta(x) = \frac{1}{2} \cdot e^{-|x-\theta|},\ -\infty < x < \infty,\ \theta \in \mathbb{R} = \Theta.$

13. (a) Let $\mathcal{P}_0$ and $\mathcal{P}_1$ be two families of distributions such that every null set of $\mathcal{P}_0$ is also a null set of $\mathcal{P}_1$, and $\mathcal{P}_0 \subseteq \mathcal{P}_1$. Show that a sufficient statistic that is complete for $\mathcal{P}_0$ is also complete for $\mathcal{P}_1$.
    (b) Let $X_1, \ldots, X_n$ be a random sample from from a population having the p.d.f. $f \in \mathcal{P}$, where $\mathcal{P}$ is the family of all the Lebesgue p.d.f.s. Show that $\underline{T}(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$ is complete. (**Hint:** In (a), take $\mathcal{P}_0$ to be the exponential family with p.d.f. $f_{\underline{\theta}} = c(\underline{\theta})e^{-\sum_{i=1}^n \theta_i \sum_{j=1}^n x_j^i - \sum_{i=1}^n x_i^{2n}}, \underline{x} \in \mathbb{R}^n, \underline{\theta} \in \mathbb{R}^n$).

MTH 418a: Inference-I

Assignment No. 3: Minimal Sufficiency and Completeness

Problem No. 1 Let the statistic $T \equiv T(\underline{X})$ be complete for the family $\{f_\theta : \theta \in \Theta\}$ of pdfs/pmfs.

Let $Y \equiv h(T)$ be a given function of T.

To show that Y is complete, let $\psi(\cdot)$ be a given function such that

$$E_\theta(\psi(Y)) = 0 \quad \forall\, \theta \in \Theta$$

$$\Rightarrow \quad E_\theta(\psi(h(T))) = 0 \quad \forall\, \theta \in \Theta$$

$$\Rightarrow \quad E_\theta(K(T)) = 0 \quad \forall\, \theta \in \Theta, \quad \text{where } K = \psi \circ h(\cdot)$$

$$\Rightarrow \quad P_\theta(\psi(h(T)) = 0) = 1 \quad \forall\, \theta \in \Theta$$

$$\Rightarrow \quad P_\theta(\psi(Y) = 0) = 1 \quad \forall\, \theta \in \Theta$$

$\Rightarrow \quad Y = h(T)$ is complete.

Problem No. 2 A statistic $T(\underline{X})$ is minimal sufficient for $\theta \in \Theta$ iff for any two sample points $\underline{x} = (x_1, \dots, x_n) \in \chi$ and $\underline{y} = (y_1, \dots, y_n) \in \chi$

$$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} \text{ is independent of } \theta \text{ iff } T(\underline{x}) = T(\underline{y}),$$

where $g_\theta(\underline{t}) = \prod_{i=1}^{n} f_\theta(t_i)$, $\underline{t} \in \chi$ is the joint pdf of $\underline{X}$.

(a)
$$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = e^{\frac{1}{2\theta^2}\left[\sum_{i=1}^{n}(y_i-\mu_0)^2 - \sum_{i=1}^{n}(x_i-\mu_0)^2\right]}$$

is independent of $\theta \in (0, \infty)$ iff $T(\underline{x}) = \sum_{i=1}^{n}(x_i-\mu_0)^2 = \sum_{i=1}^{n}(y_i-\mu_0)^2 = T(\underline{y})$

$\Rightarrow \quad T(\underline{X}) = \sum_{i=1}^{n}(X_i-\mu_0)^2$ is sufficient for $\theta \in \Theta = (0, \infty)$

Also $g_\theta(\underline{x}) = e^{-\frac{1}{2\theta^2}\sum_{i=1}^n (x_i-\mu_0)^2 - \frac{n}{2}\ln(2\pi\theta^2)}$,

$\eta = -\frac{1}{2\theta^2} \in (-\infty, 0)$ Contains an open rectangle in $\mathbb{R}$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^n (X_i-\mu_0)^2$ is Complete for $\theta \in \Theta = (0, \infty)$

Since the minimal suff. stat $T(\underline{X}) = \sum_{i=1}^n (X_i-\mu_0)^2$ is a function of $U(\underline{X}) = (\bar{X}, \sum_{i=1}^n (X_i-\bar{X})^2)$ $(T = U_2 + n(U_1-\mu_0)^2)$ $\therefore$ $\underline{U}$ is also sufficient. $\underline{U}$ is not Complete as $E_{\underline{\theta}}(U_1-\mu_0) = E_{\underline{\theta}}(\bar{X}-\mu_0) = 0 \;\forall \theta \in \Theta$ But $P_{\underline{\theta}}(\bar{X}-\mu_0 = 0) = 0 \;\forall \theta \in \Theta$

(b) $\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = e^{\frac{\theta}{\sigma_0^2}\left(\sum_{i=1}^n x_i - \sum_{i=1}^n y_i\right)} \quad e^{-\frac{1}{2\sigma_0^2}\sum_{i=1}^n x_i^2}$

is independent of θ iff $T(\underline{x}) = \sum_{i=1}^n x_i = \sum_{i=1}^n y_i = T(\underline{y})$ for $\theta \in \Theta = \mathbb{R}$

is minimal sufficient

$\Rightarrow T(\underline{X}) = \sum_{i=1}^n X_i$

Also

$g_\theta(\underline{x}) = e^{\frac{\theta}{\sigma_0^2}\sum_{i=1}^n x_i - \frac{n\theta^2}{2\sigma_0^2}} \times \left(\frac{1}{\sigma_0\sqrt{2\pi}}\right)^n e^{-\frac{\sum_{i=1}^n x_i^2}{2\sigma_0^2}}$

$\eta = \frac{\theta}{\sigma_0^2} \in \mathbb{R}$ Contains an open rectangle in $\mathbb{R}$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^n X_i$ is Complete for $\theta \in \Theta$

(C) $\frac{g_{\underline{\theta}}(\underline{x})}{g_{\underline{\theta}}(\underline{y})} = e^{\frac{\mu}{\sigma^2}\left(\sum_{i=1}^n x_i - \sum_{i=1}^n y_i\right) + \frac{1}{2\sigma^2}\left(\sum_{i=1}^n y_i^2 - \sum_{i=1}^n x_i^2\right)}$ is independent of $\underline{\theta} = (\mu, \sigma^2)$ iff $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$ and $\sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i^2$

i.e. iff $U(\underline{x}) = \left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2\right) = \left(\sum_{i=1}^n y_i, \sum_{i=1}^n y_i^2\right) = U(\underline{y})$

$\Rightarrow \hat{U}(\underline{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2\right)$ is minimal sufficient for $\underline{\theta} = (\mu, \sigma) \in \Theta$

Since $T(\underline{X}) = (\bar{X}, \sum_{i=1}^n (X_i-\bar{X})^2)$ is a 1-1 function of $U(\underline{X})$ $T(\underline{X})$ is also minimal sufficient $\left(\underline{T} = (T_1, T_2) = \left(\frac{\hat{U}_1}{n}, \hat{U}_2 - \frac{\hat{U}_1^2}{n}\right)\right.$ and $\underline{\hat{U}} = (\hat{U}_1, \hat{U}_2) = (nT_1, T_2 + nT_1^2)\Big)$

Also

$g_{\underline{\theta}}(\underline{x}) = e^{-\frac{1}{2\sigma^2}\sum x_i^2 + \frac{\theta}{\sigma^2}\sum x_i - \frac{n}{2}\ln(2\pi\sigma^2)}$

$(\eta_1, \eta_2) = \left(-\frac{1}{2\sigma^2}, \frac{\theta}{\sigma^2}\right) \in (-\infty, 0) \times \mathbb{R}$ Contains an open rectangle in $\mathbb{R}^2$

$\Rightarrow \hat{U}(\underline{X}) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2\right)$ is Complete for $\underline{\theta} = (\mu, \sigma) \in \Theta$

Thus $U(\underline{X}) = (\bar{X}, \sum_{i=1}^{n}(X_i-\bar{X})^2)$, being a function of Complete statistic $T(\underline{X})$ is also Complete.

Since the minimal sufficient statistic $T(\underline{X}) = (\bar{X}, \sum_{i=1}^{n}(X_i-\bar{X})^2)$ is not a function of $U(\underline{X}) = \bar{X}+S^2$, $U(\underline{X})$ Can not be sufficient. However $U(\underline{X}) = \bar{X}+S^2$, being a function of Complete statistic $T(\underline{X})$, is Complete.

For sample points $\underline{x}$ and $\underline{y}$

(d) $\wedge$ $$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = e^{\frac{1}{\theta}\left[\sum_{i=1}^{n}(y_i-\mu_0) - \sum_{i=1}^{n}(x_i-\mu_0)\right]}$$

is independent of θ iff $T(\underline{x}) = \sum_{i=1}^{n}(x_i-\mu_0) = \sum_{i=1}^{n}(y_i-\mu_0) = T(\underline{y})$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^{n}(X_i-\mu_0)$ is minimal sufficient

Also
$$g_\theta(\underline{x}) = e^{-\frac{1}{\theta}\sum_{i=1}^{n}(x_i-\mu_0) - n\ln\theta} \underbrace{I(x_{(1)}>\mu_0)}_{h(\underline{x})}$$

$\eta = -\frac{1}{\theta} \in (-\infty, 0)$ Contains a rectangle in $\mathbb{R}$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^{n}(X_i-\mu_0)$ is a Complete statistic for θ

As $U(\underline{X}) = \bar{X}$ is a 1-1 function of Complete and minimal sufficient statistic $T(\underline{X})$ $\left(U(\underline{X}) = \frac{T(\underline{X})}{n}+\mu_0\right)$, $U(\underline{X})$ is minimal sufficient (and hence sufficient) and Complete.

(e) $$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = e^{\frac{1}{\sigma_0}(\sum y_i - \sum x_i)} \frac{I(x_{(1)}>\theta)}{I(y_{(1)}>\theta)}$$ (here sample points $\underline{x}$ and $\underline{y} \in \mathbb{R}^n$)

is independent of θ iff $T(\underline{x}) = x_{(1)} = y_{(1)} = T(\underline{y})$

$\Rightarrow T(\underline{X}) = X_{(1)}$ is minimal sufficient for θ

As the minimal sufficient statistic $T(\underline{X}) = X_{(1)}$ is a function of $U(\underline{X}) = (X_{(1)}, \sum_{i=1}^{n}(X_i-X_{(1)}))$, $U(\underline{X})$ is sufficient.

Note that $T = X_{(1)} \sim Exp(\theta, \frac{\sigma_0}{n})$. Let $\psi(\cdot)$ be a function such that

$$E_\theta(\psi(T)) = 0, \quad \forall\, \theta \in \Theta$$

$$\Rightarrow \int_\theta^\infty \psi(t)\, \frac{n}{\sigma_0} e^{-\frac{n}{\sigma_0}(t-\theta)}\, dt = 0 \quad \forall\, \theta \in \mathbb{R}$$

$$\Rightarrow \int_\theta^\infty \psi(t)\, e^{-\frac{n}{\sigma_0}t}\, dt = 0 \quad \forall\, \theta \in \mathbb{R}$$

$$\Rightarrow \psi(\theta)\, e^{-\frac{n}{\sigma_0}\theta} = 0, \quad \text{a.e. on } \mathbb{R}$$

$$\Rightarrow \psi(t) = 0, \quad \text{a.e. on } \mathbb{R}$$

$$\Rightarrow P_\theta(\psi(T) = 0) = 1 \quad \forall\, \theta \in \Theta$$

$\Rightarrow T$ is complete.

$$E_\theta\left(\frac{1}{n-1}\sum_{i=1}^n (X_i - X_{(1)}) - \sigma_0\right) = 0, \quad \forall\, \theta \in \Theta = \mathbb{R}$$

but $P_\theta\left(\frac{1}{n-1}\sum_{i=1}^n (X_i - X_{(1)}) - \sigma_0 = 0\right) = 0 \quad \forall\, \theta \in \Theta = \mathbb{R}$

$\Rightarrow U(\underline{X}) = \left(X_{(1)}, \sum_{i=1}^n (X_i - X_{(1)})\right)$ is not complete for $\theta \in \Theta$.

(f) For sample points $\underline{x}, \underline{y} \in \mathbb{R}^n$,

$$\frac{f_\theta(\underline{x})}{f_\theta(\underline{y})} = e^{\frac{1}{\sigma}\left(\sum_{i=1}^n y_i - \sum_{i=1}^n x_i\right)} \frac{I(x_{(1)} > \mu)}{I(y_{(1)} > \mu)}$$

is independent of $\underline{\theta} = (\mu, \sigma)$ iff $x_{(1)} = y_{(1)}$ and $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$,

i.e., iff $\hat{U}(\underline{x}) = \left(x_{(1)}, \sum_{i=1}^n x_i\right) = \left(y_{(1)}, \sum_{i=1}^n y_i\right) = \hat{U}(\underline{y})$

$\Rightarrow \hat{U}(\underline{X}) = \left(X_{(1)}, \sum_{i=1}^n X_i\right)$ is minimal sufficient for $\underline{\theta}$

Since $T(\underline{X}) = \left(X_{(1)}, \sum_{i=1}^n (X_i - X_{(1)})\right)$ is a 1-1 function of $\hat{U}(\underline{X}) = \left(X_{(1)}, \sum_{i=1}^n X_i\right)$, $T(\underline{X})$ is also minimal sufficient

Let $\psi(\cdot,\cdot)$ be a function such that

$E_{\underline{\theta}}(\psi(T_1, T_2)) = 0 \quad \forall\ \underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty) = \Theta$

$$\Rightarrow \int_{\mu}^{\infty}\int_{0}^{\infty} \psi(t_1, t_2)\, \frac{n}{\sigma} e^{-\frac{n}{\sigma}(t_1-\mu)} \times \frac{1}{\Gamma(n-1)\,\sigma^{n-1}} e^{-\frac{t_2}{\sigma}} t_2^{n-2}\, dt_1\, dt_2 = 0$$

$\forall\ \mu \in \mathbb{R},\ \sigma > 0$

($T_1 = X_{(1)} \sim Exp(\mu, \frac{\sigma}{n})$ and $T_2 \sim Gamma(\sigma, n-1)$ are independently distributed)

$$\Rightarrow \int_{\mu}^{\infty}\left\{\int_{0}^{\infty}\psi(t_1, t_2)\, e^{-\frac{t_2}{\sigma}} t_2^{n-2}\, dt_2\right\} e^{-\frac{n}{\sigma}t_1}\, dt_1 = 0 \quad \forall\ \mu \in \mathbb{R},\ \sigma > 0$$

$$\Rightarrow \left(\int_{0}^{\infty}\psi(\mu, t_2)\, e^{-\frac{t_2}{\sigma}} t_2^{n-2}\, dt_2\right) e^{-\frac{n}{\sigma}\mu} = 0 \quad \forall\ \mu \in \mathbb{R},\ \sigma > 0$$

$$\Rightarrow \underbrace{\int_{0}^{\infty} e^{-\lambda t_2}\{\psi(\mu, t_2)\, t_2^{n-2}\}\, dt_2}_{} = 0, \quad \forall\ \lambda > 0,\ \mu \in \mathbb{R} \quad (\lambda = \tfrac{1}{\sigma})$$

$\hookrightarrow$ Laplace transform of $\{\psi(\mu, t_2)\, t_2^{n-2}\}$ vanishing on an interval

$\Rightarrow \psi(\mu, t_2)\, t_2^{n-2} = 0$ a.e. $\forall\ \mu \in \mathbb{R},\ t_2 > 0$

$\Rightarrow \psi(t_1, t_2) = 0$, a.e. on $\mathbb{R} \times (0, \infty)$

$\Rightarrow P_{\underline{\theta}}(\psi(T_1, T_2) = 0) = 1 \quad \forall\ \underline{\theta} \in \Theta = \mathbb{R} \times (0, \infty)$

$\Rightarrow \underline{T} = (T_1, T_2)$ is complete

(g)

$$g_\theta(\underline{x}) = e^{-\frac{1}{\theta}\sum x_i - n\alpha_0 \ln\theta}\left(\frac{1}{(\Gamma\alpha_0)^n}\prod_{i=1}^{n} I(x_i > 0)\right)$$

$\eta = -\frac{1}{\theta} \in (-\infty, 0)$ contains an open rectangle in $\mathbb{R}$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^{n} X_i$ is a complete and sufficient statistic for $\theta \in \Theta = (0, \infty)$. Consequently $T(\underline{X})$ is also minimal sufficient

(Note: Parts (a), (b), (c) can also be done in similar manner as (g))

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Problem No. 3

(a) Same as Problem 2(a)

(b) $$g_{\underline{\theta}}(\underline{x}) = e^{\theta_1 \ln(\prod_{i=1}^n x_i) + \theta_2 \ln(\prod_{i=1}^n (1-x_i)) - n \ln B(\theta_1, \theta_2)} \prod_{i=1}^n \left\{ \frac{1}{x_i(1-x_i)} I(0<x_i<1) \right\}$$

$(\eta_1, \eta_2) = (\theta_1, \theta_2) \in (0, \infty) \times (0, \infty)$ contains an open rectangle in $\mathbb{R}^2$

$\Rightarrow \hat{U}(\underline{X}) = \left(\ln(\prod_{i=1}^n X_i), \ln(\prod_{i=1}^n (1-X_i)) \right)$ is complete and sufficient

$\Rightarrow T(\underline{X}) = \left(\prod_{i=1}^n X_i, \prod_{i=1}^n (1-X_i) \right)$, being a 1-1 function of $\hat{U}(\underline{X})$,

is also complete and sufficient

$\Rightarrow T(\underline{X})$ is minimal sufficient and complete

~~(c) $\frac{f_\theta(\underline{x})}{f_\theta(\underline{y})} = e^{\frac{1}{2\theta}(\Sigma y_i^2 - \Sigma x_i^2)} e^{(\Sigma x_i - \Sigma y_i)}$ is independent of θ iff $T(\underline{x}) = \sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i^2 = T(\underline{y})$~~

~~$\Rightarrow T(\underline{X}) = \sum_{i=1}^n X_i^2$ is~~

(c) $$g_\theta(\underline{x}) = e^{-\frac{1}{2\theta}\Sigma x_i^2 - \frac{n}{2}\ln(2\pi\theta)} e^{\Sigma x_i}$$

$\eta = -\frac{1}{2\theta} \in (-\infty, 0)$ contains an open rectangle/interval in $\mathbb{R}$

$\Rightarrow T(\underline{X}) = \sum_{i=1}^n X_i^2$ is complete sufficient for θ

$\Rightarrow T(\underline{X}) = \sum_{i=1}^n X_i^2$ is minimal sufficient and complete for θ.

(d) $$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = e^{-\frac{1}{2\theta^2}(\Sigma x_i^2 - \Sigma y_i^2) + \frac{1}{\theta}(\Sigma x_i - \Sigma y_i)}$$

is independent of $\theta \in \Theta = (0, \infty)$ iff $\hat{U}(\underline{x}) = \left(\sum_{i=1}^n x_i, \sum_{i=1}^n x_i^2 \right)$
$= \left(\sum_{i=1}^n y_i, \sum_{i=1}^n y_i^2 \right) = \hat{U}(\underline{y})$

$\Rightarrow \hat{U}(\underline{X}) = \left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i^2\right)$ is minimal sufficient for θ

$\Rightarrow T(\underline{X}) = \left(\bar{X}, \sum_{i=1}^{n}(X_i-\bar{X})^2\right)$, being a 1-1 function of minimal sufficient statistic $\hat{U}(\underline{X})$ is minimal sufficient.

$E_\theta(\bar{X}^2) = \frac{\theta^2}{n} + \theta^2, \ \forall \theta>0 \Rightarrow E_\theta\left(\frac{n}{n+1}\bar{X}^2\right) = \theta^2 \quad \forall \theta>0$

$E_\theta\left(\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2\right) = \theta^2 \quad \forall \theta>0$

$\Rightarrow E_\theta\left(\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2 - \frac{n}{n+1}\bar{X}^2\right) = 0 \quad \forall \theta>0$

But $P_\theta\left(\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2 - \frac{n}{n+1}\bar{X}^2 = 0\right) = 0 \quad \forall \theta \in \Theta = (0,\infty)$

$\Rightarrow T(\underline{X}) = \left(\bar{X}, \sum_{i=1}^{n}(X_i-\bar{X})^2\right)$ is not complete for $\theta\in\Theta$

(e) For sample points $\underline{x}, \underline{y} \in \mathbb{R}^n$

$$\frac{g_{\underline{\theta}}(\underline{x})}{g_{\underline{\theta}}(\underline{y})} = \frac{I(x_{(1)}>\theta_1)\, I(x_{(n)}<\theta_2)}{I(y_{(1)}>\theta_1)\, I(y_{(n)}<\theta_2)}$$ is independent of $\underline{\theta} = (\theta_1, \theta_2)$ iff $T(\underline{x}) = (x_{(1)}, x_{(n)}) = (y_{(1)}, y_{(n)}) = T(\underline{y})$

$\Rightarrow T(\underline{X}) = (X_{(1)}, X_{(n)})$ is minimal sufficient.

Let $\psi(\cdot,\cdot)$ be a function such that

$E_{\underline{\theta}}(\psi(X_{(1)}, X_{(n)})) = 0 \quad \forall \underline{\theta}\in\Theta$

$$\Rightarrow \iint_{\theta_1<x<y<\theta_2} \psi(x,y) \frac{\lfloor n}{\lfloor n-2}\left(\frac{y-\theta_1}{\theta_2-\theta_1} - \frac{x-\theta_1}{\theta_2-\theta_1}\right)^{n-2} \times \frac{1}{(\theta_2-\theta_1)^2}\, dy\, dx = 0 \quad \forall -\infty<\theta_1<\theta_2<\infty$$

$$\Rightarrow \int_{\theta_1}^{\theta_2}\left(\int_{x}^{\theta_2}\psi(x,y)(y-x)^{n-2}\, dy\right)dx = 0, \quad \forall -\infty<\theta_1<\theta_2<\infty$$

$\Rightarrow$ for every fixed $\theta_2 \in \mathbb{R}$,

$$\int_{\theta_1}^{\theta_2}\psi(\theta_1, y)(y-\theta_1)^{n-2}\, dy = 0, \quad \forall -\infty<\theta_1<\theta_2$$

$\Rightarrow \quad \psi(\theta_1, \theta_2)(\theta_2-\theta_1)^{n-2} = 0$ a.e. on $-\infty < \theta_1 < \theta_2 < \infty$

$\Rightarrow \quad \psi(t_1, t_2) = 0$ a.e. on $-\infty < t_1 < t_2 < \infty$

$\Rightarrow \quad P_{\underset{\sim}{\theta}}(\psi(X_{(1)}, X_{(n)}) = 0) = 1 \quad \forall \ \underset{\sim}{\theta} \in \Theta$

$\Rightarrow T(\underset{\sim}{X}) = (X_{(1)}, X_{(n)})$ is complete.

(g) For sample points $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^n$

$$\frac{g_\theta(\underset{\sim}{x})}{g_\theta(\underset{\sim}{y})} = \frac{I(x_{(n)} - \frac{1}{2} \le \theta \le x_{(1)} + \frac{1}{2})}{I(y_{(n)} - \frac{1}{2} \le \theta \le y_{(1)} + \frac{1}{2})}$$

is independent of θ iff $T(\underset{\sim}{x}) = (x_{(1)}, x_{(n)}) = (y_{(1)}, y_{(n)}) = T(\underset{\sim}{y})$

$\Rightarrow \quad T(\underset{\sim}{X}) = (X_{(1)}, X_{(n)})$ is minimal sufficient.

Let $Y_i = X_i - \theta + \frac{1}{2}$, $i = 1, \ldots, n$. Then $Y_1, \ldots, Y_n$ are iid $U(0,1)$, $Y_{(1)} = X_{(1)} - \theta + \frac{1}{2}$, $Y_{(n)} = X_{(n)} - \theta + \frac{1}{2}$

$$E_\theta(X_{(n)} - X_{(1)}) = E(Y_{(n)} - Y_{(1)}) = C_n, \quad \forall \ \theta \in \mathbb{R}$$

where constant C_n does not depend on θ (as distribution of $(Y_1, \ldots, Y_n)$ does not depend on θ)

$\Rightarrow \quad E_\theta(X_{(n)} - X_{(1)} - C_n) = 0 \quad \forall \ \theta \in \Theta$

But $P_\theta(X_{(n)} - X_{(1)} - C_n = 0) = 0 \quad \forall \ \theta \in \Theta$

$\Rightarrow \quad T(\underset{\sim}{X}) = (X_{(1)}, X_{(n)})$ is not complete.

(h) $$\frac{g_\theta(\underset{\sim}{x})}{g_\theta(\underset{\sim}{y})} = \frac{\prod_{i=1}^n (\theta - x_i) \ I(x_{(n)} \le \theta)}{\prod_{i=1}^n (\theta - y_i) \ I(y_{(n)} \le \theta)}$$

is independent of θ iff $x_{(n)} = y_{(n)}$ and

$$\underbrace{\prod_{i=1}^n (\theta - x_i)}_{\text{polynomial of degree } n} = k(\underset{\sim}{x}, \underset{\sim}{y}) \underbrace{\prod_{i=1}^n (\theta - y_i)}_{\text{polynomial of degree}}, \quad \forall \ \underset{\sim}{\theta} < x_{(n)}$$

$\Rightarrow$

$\Rightarrow$ $R(\underline{x}, \underline{y}) = 1$ and

$$\prod_{i=1}^{n}(\theta - x_i) = \prod_{i=1}^{n}(\theta - y_i), \quad \forall\ \theta \le x_{(n)}$$

$\downarrow$ polynomial of degree n with roots $x_1, \dots, x_n$

$\downarrow$ polynomial of degree n with roots $y_1 \dots y_n$

$\Rightarrow \{x_1, \dots, x_n\} = \{y_1 \dots y_n\}$

$\Rightarrow T(\underline{x}) = (x_{(1)}, \dots, x_{(n)}) = (y_{(1)}, \dots, y_{(n)}) = T(\underline{y})$ is minimal sufficient for θ

$\Rightarrow T(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$

Let $Y_i = 1 - \frac{X_i}{\theta}$, $i = 1, \dots, n$. Then $Y_1, \dots, Y_n$ are iid with pdf

$$g(y) = 2y, \quad 0 < y < 1$$

$X_{(1)} = \theta(1 - Y_{(n)})$, $\quad X_{(n)} = \theta(1 - Y_{(1)})$

$E_\theta\left(\frac{X_{(1)}}{X_{(n)}}\right) = E_\theta\left(\frac{1 - Y_{(n)}}{1 - Y_{(1)}}\right) = c_n$, $\forall\ \theta > 0$ where c_n does not depend on θ (as dist. of $\underline{Y} = (Y_1, \dots, Y_n)$ does not depend on θ)

$\Rightarrow E_\theta\left(\frac{X_{(1)}}{X_{(n)}} - c_n\right) = 0 \quad \forall\ \theta > 0$

But $P_\theta\left(\frac{X_{(1)}}{X_{(n)}} - c_n = 0\right) = 0 \quad \forall\ \theta > 0$

$\Rightarrow T(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$ is not complete.

(h) $g_\theta(\underline{x}) = e^{n(\ln\theta)\bar{x} - n\theta}\left(\prod_{i=1}^{n}\frac{1}{\lfloor x_i} I(x_i \in \{0, 1, 2, \dots\})\right)$

$\eta = n\ln\theta \in (-\infty, \infty)$ contains an open interval in $\mathbb{R}$

$\Rightarrow T(\underline{X}) = \bar{X}$ is complete sufficient for $\theta \in (0,\infty)$

$\Rightarrow T(\underline{X}) = \bar{X}$ is minimal sufficient and complete for θ

(i) $g_\theta(\underline{x}) = e^{(\ln \frac{\theta}{1-\theta}) \sum_{i=1}^{n} x_i + mn \ln(1-\theta)} \left(\prod_{i=1}^{n} \binom{m}{x_i} I(x_i \in \{0, 1, \dots, m\}) \right)$

$\eta_1 = \ln(\frac{\theta}{1-\theta}) \in (-\infty, \infty) = \mathbb{R}$ contains an open rectangle in $\mathbb{R}$

$\Rightarrow$ $T(\underline{X}) = \sum_{i=1}^{n} X_i$ complete sufficient for θ

$\Rightarrow$ $T(\underline{X}) = \sum_{i=1}^{n} X_i$ is minimal sufficient and complete for θ

(j) $$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = \frac{\left(\prod_{i=1}^{n} y_i^2 I(y_i \in \{1, 2, \dots\}) \right) I(\theta \le x_{(1)}-1)}{\left(\prod_{i=1}^{n} x_i^2 I(x_i \in \{1, 2, \dots\}) \right) I(\theta \le y_{(1)}-1)}$$ iff

is independent of $\theta \in \{0, 1, 2, \dots\}$

$T(\underline{x}) = x_{(1)} = y_{(1)} = T(\underline{y})$

$\Rightarrow$ $T(\underline{X}) = X_{(1)}$ is minimal sufficient for $\theta \ge 0$

$P_\theta(X_{(1)} = x) = P_\theta(X_{(1)} > x-1) - P_\theta(X_{(1)} > x)$ $\quad x = \theta+1, \theta+2, \dots$

$= [C(\theta)]^n \left[\left(\sum_{j=x}^{\infty} \frac{1}{j^2} \right)^n - \left(\sum_{j=x+1}^{\infty} \frac{1}{j^2} \right)^n \right]$, $\quad \theta \in \{0, 1, 3, \dots\}$

Let $\psi(\cdot)$ be a function such that

$E_\theta(\psi(X_{(1)})) = 0$, $\quad \forall\ \theta = 0, 1, 2, \dots$

$\Rightarrow$ $[C(\theta)]^n \sum_{x=\theta+1}^{\infty} \psi(x) \left[\left(\sum_{j=x}^{\infty} \frac{1}{j^2} \right)^n - \left(\sum_{j=x+1}^{\infty} \frac{1}{j^2} \right)^n \right] = 0$

$\forall\ \theta = 0, 1, 2, \dots$

$\Rightarrow$ $\sum_{x=\theta+1}^{\infty} \psi(x) \left[\left(\sum_{j=x}^{\infty} \frac{1}{j^2} \right)^n - \left(\sum_{j=x+1}^{\infty} \frac{1}{j^2} \right)^n \right] = 0$ $\quad \forall\ \theta = 0, 1, 2, \dots$ ---(A)

and $\sum_{x=\theta+2}^{\infty} \psi(x) \left[\left(\sum_{j=x}^{\infty} \frac{1}{j^2} \right)^n - \left(\sum_{j=x+1}^{\infty} \frac{1}{j^2} \right)^n \right] = 0$ $\quad \forall\ \theta = 1, 2, \dots$

Subtracting we get

$\psi(\theta+1) \left[\left(\sum_{j=\theta+1}^{\infty} \frac{1}{j^2} \right)^n - \left(\sum_{j=\theta+2}^{\infty} \frac{1}{j^2} \right)^n \right] = 0$ $\quad \forall\ \theta = 1, 2, \dots$

$\Rightarrow$ $\psi(\theta+1) = 0$, $\quad \forall\ \theta = 1, 2, \dots$

$\Rightarrow$ $\psi(\theta+1) = 0$, for $\theta = 0$ (from (A))

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Thus $\psi(x) = 0, \ \forall x = 1, 2, \ldots$

$\Rightarrow P_\theta(\psi(X_{(1)}) = 0) = 1, \quad \forall \theta \in \{0, 1, \ldots\} = \Theta$

(as $P_\theta(X_{(1)} \in \{1, 2, \ldots\}) = 1, \ \forall \theta \in \{0, 1, \ldots\}$)

$\Rightarrow T(\underline{X}) = X_{(1)}$ is complete.

Problem No. 4

(a) See Problem 2(a)

(b) $U(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$ is not minimal sufficient as it is not a function of sufficient statistic $T(\underline{X}) = \left(\prod_{i=1}^{n} X_i, \prod_{i=1}^{n} (1-X_i)\right)$

Clearly $U(\underline{X})$ is sufficient (as sufficient statistic $T(\underline{X}) = \left(\prod_{i=1}^{n} X_{(i)}, \prod_{i=1}^{n} (1-X_{(i)})\right)$ is a function of $U(\underline{X})$). If $U(\underline{X})$ was also complete then being complete-sufficient it will be minimal sufficient, which is not true.

Hence $U(\underline{X})$ is not complete.

(c) $\frac{dU}{dT} = 30T + 8 > 0$, as $P_\theta(T>0) = 1$, $\forall \theta > 0$

$\Rightarrow$ U is a 1-1 function of minimal sufficient and complete statistic $T(\underline{X}) = \sum_{i=1}^{n} X_i^2$

$\Rightarrow$ $U(\underline{X})$ is minimal sufficient and complete.

(d) Note that $\sum_{i=1}^{n} (X_i - 1)^2 = \sum_{i=1}^{n} (X_i - \bar{X})^2 + n(\bar{X}-1)^2$

and $\sum_{i=1}^{n} (X_i - \bar{X})^2 = \sum_{i=1}^{n} (X_i - 1)^2 - n(\bar{X}-1)^2$

Thus $U(\underline{X}) = \left(\bar{X}, \sum_{i=1}^{n} (X_i - \bar{X})^2\right)$, being an 1-1 function of minimal sufficient statistic $T(\underline{X}) = \left(\bar{X}, \sum_{i=1}^{n} (X_i - \bar{X})^2\right)$, is minimal sufficient. Since $\underline{T}$ is not complete, and $\underline{T}$ is a function of $\underline{U}$, U will also be not complete.

(e) $U(\underline{X}) = (2X_{(n)} - X_{(1)}, X_{(n)} + 3X_{(1)})$ is a 1-1 function of complete-sufficient statistic $T(\underline{X}) = (X_{(1)}, X_{(n)})$. Thus

$\Rightarrow$ $U(\underline{X})$ is complete-sufficient

$\Rightarrow$ $U(\underline{X})$ is minimal sufficient and complete

(f) As $U(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$ is not a function of sufficient statistic $T(\underline{X}) = (X_{(1)}, X_{(n)})$, $U(\underline{X})$ is not minimal sufficient. Clearly $U(\underline{X})$ is sufficient (as sufficient statistic $T(\underline{X})$ is a function of $U(\underline{X})$). Thus if $U(\underline{X})$ is complete then it is complete-sufficient and hence minimal sufficient, which is not true.

Thus $U(\underline{X})$ is not complete

(g) $U(\underline{X}) = (\prod_{i=1}^{n} X_{(i)}, X_{(2)} + X_{(3)}, 2X_{(2)} + 3X_{(3)}, X_{(4)}, \ldots, X_{(n)})$ is a 1-1 function of minimal sufficient statistic $T(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$. Hence $U(\underline{X})$ is minimal sufficient

Since $T(\underline{X})$ is a function of $U(\underline{X})$, and $T(\underline{X})$ is not complete, $U(\underline{X})$ cannot be complete.

(h) Note that $\bar{X} \geq 0$, w.p. 1

$$\frac{dU}{d\bar{X}} = 4\bar{X} + 3 > 0 \quad \text{w.p. 1}$$

$\Rightarrow$ U is a 1-1 function of complete and minimal sufficient statistic $\bar{X}$

$\Rightarrow$ U is minimal sufficient and complete

(i) $\bar{X} \geq 0$ w.p. 1 and $\bar{X} = \frac{T(\underline{X})}{n}$ is minimal sufficient and complete ($\bar{X}$ is a 1-1 function of minimal sufficient and complete statistic $T(\underline{X}) = \sum_{i=1}^{n} X_i$)

$$\frac{dU}{d\bar{X}} = 5\bar{X}^4 \geq 0 \quad \text{w.p.}$$

$\Rightarrow$ U is a 1-1 function of minimal sufficient and complete statistic $\bar{X}$

$\Rightarrow$ U is minimal sufficient and complete

(j) $X_{(1)} > 0$ w.p. 1

$$\frac{dU}{dX_{(1)}} = 18e^{2X_{(1)}} - 17e^{X_{(1)}} = e^{X_{(1)}}(18e^{X_{(1)}} - 17) > 0 \quad \text{w.p. 1}$$

$\Rightarrow$ U is a 1-1 function of minimal sufficient statistic $T(\underline{X}) = X_{(1)}$

$\Rightarrow$ $U(\underline{X})$ is minimal sufficient

Since $U(\underline{X})$ is a function of complete statistic $T(\underline{X})$, $U(\underline{X})$ is complete.

Problem No. 5 X_1, X_2 are iid with p.m.f.

$$f_\theta(x) = \begin{cases} \frac{1}{3}, & x = \theta, \theta+1, \theta+2 \\ 0 & \text{otherwise} \end{cases},$$

$\theta \in \Theta = \{0, 1, 2, \dots\}$.

The joint p.d.f. of $\underline{X} = (X_1, X_2)$ is

$$g_\theta(x_1, x_2) = \frac{1}{9} \prod_{i=1}^{n} I(x_i \in \{\theta, \theta+1, \theta+2\})$$

$$= \frac{1}{9} I(x_{(2)} - 2 \le \theta \le x_{(1)}) \prod_{i=1}^{2} I\{x_i \in \{0, 1, 2, \dots\}\},$$

$\theta = 0, 1, 2, \dots$

For any sample points (x_1, x_2) and (y_1, y_2) belonging to $\{0, 1, 2, \dots\}^2$,

$$\frac{g_\theta(x_1, x_2)}{g_\theta(y_1, y_2)} = \frac{I(x_{(2)} - 2 \le \theta \le x_{(1)})}{I(y_{(2)} - 2 \le \theta \le y_{(1)})}$$

is independent of θ iff $U(\underline{x}) = (x_{(1)}, x_{(2)}) = (y_{(1)}, y_{(2)}) = U(\underline{y})$

$\Rightarrow$ $U(\underline{X}) = (X_{(1)}, X_{(2)})$ is minimal sufficient

$\Rightarrow$ $T(\underline{X}) = \left(\frac{X_{(1)} + X_{(2)}}{2}, \frac{X_{(2)} - X_{(1)}}{2} \right)$, being a 1-1 function of minimal sufficient statistic $U(\underline{X})$, is minimal sufficient

Note that

$P_\theta(X_{(1)} = \theta+2) = P_\theta(X_1 = X_2 = \theta+2) = \frac{1}{9}$

$P_\theta(X_{(1)} = \theta+1) = P(X_1 = \theta+1, X_2 = \theta+1) + 2P(X_1 = \theta+1, X_2 = \theta+2) = \frac{3}{9}$

$P_\theta(X_{(1)} = \theta) = 1 - (\frac{1}{9} + \frac{3}{9}) = 5/9$

$P_\theta(X_{(2)} = \theta) = P_\theta(X_1 = X_2 = \theta) = \frac{1}{9}$

$P_\theta(X_{(2)} = \theta+1) = P_\theta(X_1 = X_2 = \theta+1) + 2P_\theta(X_1 = \theta,\ X_2 = \theta+1) = \frac{3}{9}$

$P_\theta(X_{(2)} = \theta+2) = 1 - (\frac{1}{9} + \frac{3}{9}) = 5/9$

$E_\theta(X_{(1)}) = \frac{1}{9}(\theta+2) + \frac{3}{9}(\theta+1) + \frac{5}{9}\theta = \theta + \frac{5}{9}$

$E_\theta(X_{(2)}) = \frac{\theta}{9} + \frac{3(\theta+1)}{9} + \frac{5(\theta+2)}{9} = \theta + \frac{13}{9}$

$E_\theta(X_{(2)} - X_{(1)} - \frac{8}{9}) = 0, \quad \forall\ \theta = 0, 1, 2, \dots$

But $P_\theta(\underbrace{X_{(2)} - X_{(1)}}_{\text{takes integer values only}} - \frac{8}{9} = 0) = 0 \quad \forall\ \theta = 0, 1, 2, \dots$

$\Rightarrow U(\underline{X}) = (X_{(1)}, X_{(2)})$ is not-complete

$\Rightarrow T(\underline{X}) = (\frac{X_{(1)} + X_{(2)}}{2}, X_{(2)} - X_{(1)})$ is not complete (as $U(\underline{X})$ is a function of $T(\underline{X})$).

$\boxed{\text{Problem No. 6}}$ Let $\psi(\cdot)$ be a function such that

$$E_\theta(\psi(X)) = 0, \quad \forall\ \theta \in (0, 1)$$

$\Rightarrow \psi(-1)\theta + (1-\theta)^2 \sum_{x=0}^{\infty} \psi(x)\theta^x = 0, \quad \forall\ \theta \in (0,1)$

$\Rightarrow \psi(-1)\theta(1-\theta)^{-2} + \sum_{x=0}^{\infty} \psi(x)\theta^x = 0, \quad \forall\ \theta \in (0,1)$

$\Rightarrow \psi(-1)\theta(1 + 2\theta + 3\theta^2 + \dots) + \sum_{x=0}^{\infty} \psi(x)\theta^x = 0, \quad \forall\ \theta \in (0,1)$

$\Rightarrow \psi(-1)\sum_{x=1}^{\infty} x\theta^x + \sum_{x=0}^{\infty} \psi(x)\theta^x = 0, \quad \forall\ \theta \in (0,1)$

$\Rightarrow \psi(0) = 0$ and $\psi(-1)x + \psi(x) = 0, \quad x = 1, 2, \dots$

$\Rightarrow \psi(x) = -x\,\psi(-1), \quad x = 0, 1, 2, \dots \quad \dots (A)$

Case I $\psi(\cdot)$ is bounded

In this case we must have $\psi(-1) = 0$ as then otherwise $\psi(\cdot)$ can not be bounded due to (A)

$\Rightarrow \psi(x) = 0, \quad x = 0, 1, 2, \dots$

$\Rightarrow P_\theta(\psi(X) = 0) = 1 \quad \forall\ \theta \in \{0, 1, 2, \dots\} = \Theta$

$\Rightarrow \{b_\theta : \theta \in \Theta\}$ is boundedly complete

Case II: $\psi(\cdot)$ is unbounded

Let $\psi(-1) = -1$, so that $\psi(x) = x$, $x = 0, 1, 2, \ldots$

Then

$$E_\theta(\psi(X)) = -\theta + (1-\theta)^2 \sum_{x=0}^{\infty} x\theta^x$$

$$= (1-\theta)^{-2} \left[\sum_{x=1}^{\infty} x\theta^x - \theta(1-\theta)^{-2} \right) = 0$$

But $P_\theta(\psi(X) = 0) = 0$ $\forall\ \theta \in (0, 1)$

Thus $\{f_\theta : \theta \in \Theta\}$ is not - complete.

$\boxed{\text{Problem No. 7}}$ (a) $\Theta = \{1, 2, \ldots\}$, $f_\theta(x) = \begin{cases} \frac{1}{\theta}, & x = 1, 2, \ldots, \theta \\ 0 & \text{otherwise} \end{cases}$, $\theta \in \Theta$

Let $\psi(\cdot)$ be a function such that

$$E_\theta(\psi(X)) = 0, \quad \forall\ \theta = 1, 2, 3, \ldots$$

$$\Rightarrow \sum_{x=1}^{\theta} \frac{\psi(x)}{\theta} = 0, \quad \forall\ \theta = 1, 2, 3, \ldots$$

$$\Rightarrow \sum_{x=1}^{\theta} \psi(x) = 0, \quad \forall\ \theta = 1, 2, 3, \ldots \quad \ldots (A)$$

$$\sum_{x=1}^{\theta+1} \psi(x) = 0, \quad \forall\ \theta = 1, 2, 3, \ldots \quad (B)$$

$\Rightarrow \psi(1) = 0$ and (On substituting B−A)

$\Rightarrow \psi(1) = 0$ and $\psi(\theta+1) = 0$, $\forall\ \theta = 1, 2, 3, \ldots$

$\Rightarrow \psi(x) = 0$, $x = 1, 2, \ldots$

$\Rightarrow P_\theta(\psi(X) = 0) = 1$, $\forall\ \theta = 1, 2, 3, \ldots$

$\Rightarrow \{f_\theta : f_\theta \in \Theta\}$ is Complete

(b) $\Theta = \{1, 2, \ldots, n_0 - 1, n_0 + 1, n_0 + 2, \ldots\}$

Let $\psi(\cdot)$ be such that

$$E_\theta(\psi(X)) = 0, \quad \forall\ \theta \in \Theta$$

$$\Rightarrow \sum_{x=1}^{\theta} \psi(x) = 0 \quad \forall\ \theta = 1, 2, \ldots, n_0 - 1, n_0 + 1, \ldots \quad (A)$$

$$\sum_{x=1}^{\theta-1} \psi(x) = 0, \qquad \theta = 2, 3, \ldots, m_0, m_0+2, \ldots \ldots \quad (B)$$

Subtracting, we get (A - B)

$$\psi(\theta) = 0 \quad \theta = 2, 3, \ldots, \quad \theta \neq m_0, m_0+1$$

$$\psi(1) = 0, \quad \text{by (A)} \quad (\text{with } \theta = 1)$$

For $\theta = m_0+1$, we get from (A)

$$\psi(m_0) + \psi(m_0+1) = 0$$

Thus we have

$$\psi(k) = 0, \quad k \in \{1, 2, \ldots\} - \{m_0, m_0+1\}$$

$$\psi(m_0) + \psi(m_0+1) = 0.$$

Let $\psi(\cdot)$ be such that

$$\psi(x) = \begin{cases} 0, & x = 1, 2, 3, \ldots, \quad x \neq m_0, m_0+1 \\ -1, & x = m_0 \\ 1, & x = m_0 \end{cases}$$

Then

$$E_\theta(\psi(X)) = \sum_{x=1}^{\theta} \frac{\psi(x)}{\theta} = \begin{cases} 0, & \theta = 1, 3, \ldots, m_0-1 \\ \dfrac{\psi(m_0) + \psi(m_0+1)}{\theta}, & \theta \in \{m_0+1, m_0+2, \ldots\} \end{cases}$$

$$\text{But} \quad P_{\theta=m_0+1}(\psi(X) = 0) = 1 - \overbrace{P_{\theta=m_0+1}(X = m_0)}^{=0} - P_{\theta=m_0+1}(X = m_0+1)$$

$$= 1 - \frac{2}{m} \neq 1$$

$\Rightarrow \{f_\theta : \theta \in \Theta - \{m_0\}\}$ is not Complete.

(c) For $k = 1, 2, \ldots, \theta$

$$P_\theta(X_{(n)} = k) = P_\theta(X_{(n)} \leq k) - P_\theta(X_{(n)} \leq k-1)$$

$$= \left(\frac{k}{\theta}\right)^n - \left(\frac{k-1}{\theta}\right)^n$$

Let $\psi(\cdot)$ be such that

$$E_\theta(\psi(X_{(n)})) = 0, \quad \forall \theta = 1, 2, \ldots$$

$$\sum_{k=1}^{\theta} \psi(k)\left[k^n-(k-1)^n\right]=0, \quad \forall\ \theta=1,2,\ldots \quad (A)$$

$$\sum_{k=1}^{\theta-1} \psi(k)\left[k^n-(k-1)^n\right]=0, \quad \forall\ \theta=2,3,\ldots \quad (B)$$

For $\theta=1$, (A) gives $\psi(1)=0$.

For $\theta=2,3,\ldots$, (A)−(B) gives $\psi(\theta)=0$. Thus $\psi(x)=0$, $x=1,2,\ldots$. Consequently

$$P_\theta(\psi(X_{(n)})=0)=1, \quad \forall\ \theta=1,2,3,\ldots$$

$\Rightarrow$ $T(\underline{X})=X_{(n)}$ is complete.

$\boxed{\text{Problem No. 8}}$ The jt. p.d.f. of $(X_{(1)},\ldots,X_{(n)})$ is

$$g(x_1,\ldots,x_n)=\underline{n}\left(\frac{1}{\sigma}\right)^n e^{-\frac{1}{\sigma}\sum_{i=1}^{n}(x_i-\mu)},$$

$$\mu<x_1<x_2<\cdots<x_n<\infty$$

Define $Y_1=X_{(1)}$, $Y_2=X_{(2)}-X_{(1)}$, $Y_3=X_{(3)}-X_{(1)},\ldots$, $Y_i=X_{(i)}-X_{(i-1)}$, $i=1,\ldots,n$,
$(X_{(0)}=0)$.

Making the transformation

$$\begin{aligned} y_1&=x_1\\ y_2&=x_2-x_1\\ y_3&=x_3-x_1\\ &\vdots\\ y_n&=x_n-x_{n-1}\end{aligned} \quad\Rightarrow\quad \begin{aligned} x_1&=y_1\\ x_2&=y_1+y_2\\ x_3&=y_1+y_3\\ &\vdots\\ x_n&=y_1+y_n\end{aligned} \quad\Rightarrow\quad J=1,$$

$\mu<x_1<x_2<x_3<\cdots<x_n<\infty$

$\Rightarrow \mu<y_1<y_1+y_2<y_1+y_3,<\cdots<y_1+y_n$

$\Rightarrow y_1>\mu,\ 0<y_2<y_3<\cdots<y_n<\infty$

$$\sum_{i=1}^{n} x_i = ny_1+\sum_{i=2}^{n} y_i$$

The joint p.d.f. of $\underline{Y}=(Y_1,\ldots,Y_n)$ is

$$f_{Y_1,\ldots,Y_n}(y_1,\ldots,y_n)=\frac{\underline{n}}{\sigma^n}e^{-\frac{1}{\sigma}\left(ny_1+\sum_{i=2}^{n}y_i-n\mu\right)} I(y_1>\mu,\ 0<y_2<y_3<\cdots<y_n<\infty)$$

$$=\left\{\frac{n}{\sigma}e^{-\frac{n}{\sigma}(y_1-\mu)} I(y_1>\mu)\right\}\times\left\{\frac{\underline{n-1}}{\sigma^{n-1}}e^{-\frac{1}{\sigma}\sum_{i=2}^{n}y_i} I(0<y_2<y_3<\cdots<y_n)\right\}$$

$\mathrm{Exp}(\mu,\frac{\sigma}{n})$ $\Rightarrow$ pdf with term involving y_1 alone

dist. of O.S based on random sample of size $n-1$ from $\mathrm{Exp}(0,1)$

Let $E \sim Exp(\mu, \frac{\sigma}{n})$

$E_1 \dots E_{n-1}$ are iid $Exp(0, \sigma)$

$E_{(1)} \dots, E_{(n-1)}$ are O.S. of $E_1 \dots, E_{n-1}$. Then

$$\left.\begin{array}{l} Y_1 \stackrel{d}{=} E \\ (Y_2, \dots, Y_n) \stackrel{d}{=} (E_{(1)} \dots, E_{(n-1)}) \end{array}\right\} \text{Independent.}$$

$$\Rightarrow \left.\begin{array}{l} Y_1 = X_{(1)} \sim Exp(\mu, \frac{\sigma}{n}) \\ \sum_{i=1}^{n} (X_i - X_{(1)}) = \sum_{i=2}^{n} Y_i \sim Gamma(\sigma, n-1) \end{array}\right\} \text{Independent}$$

Problem 9 $\mathcal{P}_0 \subseteq \mathcal{P}$, T is minimal sufficient for $\mathcal{P}_0$ and sufficient for $\mathcal{P}$; densities in $\mathcal{P}$ have the same support (given)

<u>Claim:</u> T is minimal sufficient for $\mathcal{P}$

Let U be a given sufficient statistic for $\mathcal{P}$. We will show that T is a function of U, a.e. $\mathcal{P}$.

$\Rightarrow$ U is sufficient for $\mathcal{P}_0$ (as $\mathcal{P}_0 \subseteq \mathcal{P}$)

$\Rightarrow$ T is a function of U a.s. $\mathcal{P}_0$ (T is minimal suff. for $\mathcal{P}_0$)

$\Rightarrow$ T is a function of U a.s. $\mathcal{P}$ (as all distributions in $\mathcal{P}$ have the same support)

$\Rightarrow$ T is minimal sufficient for $\mathcal{P}$

Problem No. 10 (a) $\mathcal{P} = \{f_0, f_1, \dots, f_k\}$; each f_i has the same support (given)

<u>Claim:</u> $T(\underline{x}) = \left(\frac{f_1(\underline{x})}{f_0(\underline{x})}, \dots, \frac{f_k(\underline{x})}{f_0(\underline{x})} \right)$ is minimal sufficient.

Note that

$$f_0(x) = f_0(x) \quad \text{a.s. } \mathcal{P}$$

$$f_i(x) = \frac{f_i(x)}{f_0(x)} f_0(x) \quad \text{a.s. } \mathcal{P}, \quad i = 1, \dots, k$$

Taking $h(x) = f_0(x)$ a.s. $\mathcal{P}$ and $\psi(i, (T_1, \dots, T_k)) = T_i$, $i = 1, \dots, k$, and $\psi(0, (T_1, \dots, T_k)) = 1$, we can write

$$f_i(x) = \psi(i, T(\underline{x})) h(\underline{x}) \quad \text{a.s. } \mathcal{P}$$

$\Rightarrow$ $T(\underline{x})$ is sufficient (by factorization theorem)

Let $U(\underline{X})$ be any other sufficient statistic. By factorization theorem, there exist functions $M(i, U(\underline{x}))$ and $K(\underline{x})$ such that

$$f_i(\underline{x}) = M(i, U(\underline{x}))\, K(\underline{x}) \quad \text{a.s. } \mathcal{P}, \qquad i = 0, 1, \ldots, k$$

$$\Rightarrow T_i(\underline{x}) = \frac{f_i(\underline{x})}{f_0(\underline{x})} = \frac{M(i, U(\underline{x}))}{M(0, U(\underline{x}))}, \quad i = 0, 1, \ldots, k \quad \text{a.s. } \mathcal{P}$$

$$\Rightarrow T(\underline{x}) = (T_1(\underline{x}), \ldots, T_k(\underline{x}))$$

$$= \left(\frac{M(1, U(\underline{x}))}{M(0, U(\underline{x}))}, \ldots, \frac{M(k, U(\underline{x}))}{M(0, U(\underline{x}))} \right)$$

$\longrightarrow$ a function of $U(\underline{x})$

$\Rightarrow T(\underline{X})$ is minimal sufficient for $\mathcal{P}$

(b) Here support of $\mathcal{P}$ is $[-1, 1]$

$$T(\underline{x}) = \left(\frac{f_1(x)}{f_0(x)}, \frac{f_2(x)}{f_0(x)} \right) = \begin{cases} (0, 0), & \text{if } -1 < x < 0 \\ (0, 0) & \text{if } 0 < x < 1 \end{cases}$$

$$P_{f_1}(X \le x \mid T(x) = (0, 0)) = \begin{cases} 0, & x < 0 \\ x, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

$$P_{f_2}(X \le x \mid T(x) = (0, 0)) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

$\Rightarrow P_{f_i}(X \le x \mid T(x) = \underline{t})$ depends on f_i

$\Rightarrow T(\underline{X})$ is not sufficient for $\mathcal{P}$.

A statistic $U(x)$ will be minimal sufficient iff

$U(x) = U(y) \iff$ for every $i \in \{0, 1, 2\}$ $\quad f_i(x) = k(x, y)\, f_i(y)$ $\quad \cdots$ (I)

For $x \in (-1, 0)$ and $y \in (0, 1)$, (I) can never hold (we get $1 = 0$ for $i = 0$)

For $x, y \in (-1, 0)$, we get, $1 = k(x, y)$, for $i = 0$, and $0 = 0$ for $i = 1, 2$ $\longrightarrow$ satisfied for $k(x, y) = 1$

For $x, y \in (0, 1)$, we get $0 = 0$, for $i = 0$, $1 = k(x, y)$, for $i = 1$

$2x = k(x, y) \times 2y$, for $i = 2$

$\longrightarrow$ Choice of k depends on i.

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Thus a minimal sufficient statistic for P is

$$T_a(x) = \begin{cases} a, & -1 < x < 0 \\ x, & 0 < x < 1 \end{cases}$$

where 'a' is any arbitrary constant.

$\boxed{\text{Problem No. 11}}$ (a) $\theta \in \Theta = (0, 1)$

$$f_\theta(x) = \begin{cases} \frac{1}{4}, & x = 1, 2 \\ \frac{1+\theta}{4}, & x = 3 \\ \frac{1-\theta}{4}, & x = 4 \end{cases}$$

A statistic $T(x)$ is minimal sufficient iff

$T(x) = T(y) \iff \frac{f_\theta(x)}{f_\theta(y)}$ is independent of $\theta \in (0,1)$, $x, y \in \{1, 2, 3, 4\}$

$\iff x, y \in \{1, 2\}$ or $x, y \in \{3\}$, or $x, y \in \{4\}$.

A typical sufficient statistic is

$$T(x) = \begin{cases} a, & x = 1, 2 \\ b, & x = 3 \\ c, & x = 4 \end{cases}$$

where a, b and c are distinct constants.

Let $\psi(\cdot)$ be such that $E_\theta(\psi(x)) = 0$, $\forall \theta \in (0, 1)$

$\Rightarrow \frac{\psi(a)}{4} + \frac{\psi(a)}{4} + \frac{1+\theta}{4}\psi(b) + \frac{1-\theta}{4}\psi(c) = 0$, $\forall \theta \in (0,1)$

$\Rightarrow \psi(b) = \psi(c)$ and $2\psi(a) + \psi(b) + \psi(c) = 0$

$\Rightarrow \psi(b) = \psi(c)$ and $\psi(a) = -\psi(b)$.

Choose $\psi(a) = -1$, $\psi(b) = \psi(c) = 1$.

Then $E_\theta(\psi(T(x))) = 0$ $\forall \theta \in (0,1)$

But $P_\theta(\psi(T(x)) = 0) = 0$ $\forall \theta \in (0,1)$

$\Rightarrow T(\underline{x})$ is not complete

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(b) $\mathcal{X} = \{-1, 0, 1\}$, $\Theta = \{1, 2, 3\}$

$\theta \backslash x$	-1	0	1
1	0.4	0.2	0.4
2	0.6	0.3	0.1
3	0.2	0.1	0.7

For sample points $x, y \in \{-1, 0, 1\}$

$$\frac{f_1(x)}{f_1(y)} = \begin{cases} 1, & x = y \in \{-1, 0, 1\} \\ 2 & x = -1, \ y = 0 \\ 1, & x = -1, \ y = 1 \\ \frac{1}{2} & x = 0, \ y = 1 \end{cases}$$

$$\frac{f_2(x)}{f_2(y)} = \begin{cases} 1, & x = y \in \{-1, 0, 1\} \\ 2 & x = -1, \ y = 0 \\ 6 & x = -1, \ y = 1 \\ 3 & x = 0, \ y = 1 \end{cases}$$

$$\frac{f_3(x)}{f_3(y)} = \begin{cases} 1, & x = y \in \{-1, 0, 1\} \\ 2 & x = -1, \ y = 0 \\ 2/7 & x = -1, \ y = 1 \\ 1/7 & x = 0, \ y = 1 \end{cases}$$

Thus $\frac{f_i(x)}{f_i(y)}$ is independent of $x = y$ or $x, y \in \{-1, 0\}$

A typical minimal sufficient statistic is

$$T(x) = \begin{cases} a & x = -1, 0 \\ b & 1 \end{cases}$$

where a and b are distinct constants

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Let $\psi(\cdot)$ be a given function such that

$$E_f(\psi(T(X))) = 0, \quad \forall f \in \{f_1, f_2, f_3\}$$

$$\Rightarrow \quad 0.6\,\psi(a) + 0.4\,\psi(b) = 0$$
$$0.9\,\psi(a) + 0.1\,\psi(b) = 0$$
$$0.3\,\psi(a) + 0.7\,\psi(b) = 0$$

$$\Rightarrow \quad \psi(a) = \psi(b) = 0$$

$$\Rightarrow \quad P_\theta(\psi(T(X)) = 0) = 0, \quad \forall \theta \in \{1, 2, 3\}.$$

$$\Rightarrow \quad T(X) \text{ is complete.}$$

$\boxed{\text{Problem No. 12}}$ (a) For $\underline{x}, \underline{y} \in \mathbb{R}^n$

$$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = \frac{\prod_{i=1}^n [1+(y_i-\theta)^2]}{\prod_{i=1}^n [1+(x_i-\theta)^2]} \text{ is independent of } \theta \in \mathbb{R}$$

$$\Leftrightarrow \underbrace{\prod_{i=1}^n [1+(y_i-\theta)^2]}_{\text{polynomial of degree } 2n \text{ in } \theta} = \underbrace{k(\underline{x}, \underline{y}) \prod_{i=1}^n [1+(x_i-\theta)^2]}_{\text{polynomial of degree } 2n \text{ in } \theta}, \quad \forall \theta \in \mathbb{R}$$

Comparing the coefficients of θ^{2n} on both side we get $k(\underline{x}, \underline{y}) \equiv 1$

$$\Rightarrow \underbrace{\prod_{i=1}^n [1+(y_i-\theta)^2]}_{\substack{\text{polynomial of degree } 2n \\ \text{in } \theta \text{ with roots} \\ \{y_i \pm \sqrt{i},\ i=1\ldots n\}}} = \underbrace{\prod_{i=1}^n [1+(x_i-\theta)^2]}_{\substack{\text{polynomial of degree } 2n \text{ with} \\ \text{roots } \{x_i \pm \sqrt{i},\ i=1\ldots n\}}}, \quad \forall \theta \in \mathbb{R}$$

Thus

$$\{y_i \pm \sqrt{i},\ i=1\ldots n\} = \{x_i \pm \sqrt{i},\ i=1\ldots n\}$$

$$\Leftrightarrow \{y_1, \ldots, y_n\} = \{x_1, \ldots, x_n\}$$

$$\Leftrightarrow y_{(i)} = x_{(i)}, \quad i=1\ldots n$$

$$\Rightarrow T(\underline{X}) = (X_{(1)}, \ldots, X_{(n)}) \text{ is minimal sufficient.}$$

Note that $X_{(n)} - X_{(1)}$ is ancillary. Let

$P_\theta(X_{(n)} - X_{(1)} > 1) = c_n \in (0,1)$
↳ does not depend on θ

Define

$$\psi(X_{(1)}, \dots, X_{(n)}) = \begin{cases} 1 - c_n, & \text{if } X_{(n)} - X_{(1)} > 1 \\ -c_n & \text{if } X_{(n)} - X_{(1)} < 1 \end{cases}$$

Then

$E_\theta[\psi(X_{(1)}, \dots, X_{(n)})] = (1-c_n)c_n - c_n(1-c_n) = 0, \quad \forall \theta \in \Theta$

But $P_\theta(\psi(X_{(1)}, \dots, X_{(n)}) = 0) = 0 \quad \forall \theta \in \Theta$

$\Rightarrow (X_{(1)}, \dots, X_{(n)})$ is not complete.

(b) For $\underline{x}, \underline{y} \in \mathbb{R}^n$

$$\frac{g_\theta(\underline{x})}{g_\theta(\underline{y})} = e^{\sum_{i=1}^n |y_i - \theta| - \sum_{i=1}^n |x_i - \theta|}$$

is independent of θ iff

$\sum_{i=1}^n |y_{(i)} - \theta| - \sum_{i=1}^n |x_{(i)} - \theta| = k(\underline{x}, \underline{y})$, for some fn $k(\cdot,\cdot)$
$\forall \theta \in \mathbb{R}$

By choosing $\theta < \min\{x_{(1)}, y_{(1)}\}$ we get $k(\underline{x}, \underline{y}) = \sum_{i=1}^n y_{(i)} - \sum_{i=1}^n x_{(i)}$

By choosing $\theta > \max\{x_{(n)}, y_{(n)}\}$, we get $k(\underline{x}, \underline{y}) = \sum_{i=1}^n x_{(i)} - \sum_{i=1}^n y_{(i)}$

$\Rightarrow k(\underline{x}, \underline{y}) = 0$

and $\sum_{i=1}^n |x_{(i)} - \theta| = \sum_{i=1}^n |y_{(i)} - \theta|, \quad \forall \theta \in \mathbb{R}$

<u>Claim I</u> $x_{(1)} = y_{(1)}$

On contrary, WLOG let $x_{(1)} < y_{(1)}$. Then by choosing $x_{(1)} < \theta < \min\{y_{(1)}, x_{(2)}\}$ we get

$\theta - x_{(1)} + \sum_{i=2}^n (x_{(i)} - \theta) = \sum_{i=1}^n (y_{(i)} - \theta), \quad \forall \theta \in (x_{(1)}, \min\{y_{(1)}, x_{(2)}\})$

$$\Rightarrow \quad 2\theta = \sum_{i=1}^{n} y_{(i)} + x_{(1)} - \sum_{i=2}^{n} x_{(i)}, \quad \forall\ \theta \in (x_{(1)}, \min\{y_{(1)}, x_{(2)}\})$$

$\rightarrow$ Can not hold

$$\Rightarrow \quad x_{(1)} = y_{(1)} \quad \Rightarrow \quad \sum_{i=2}^{n} |x_{(i)} - \theta| = \sum_{i=2}^{n} |y_{(i)} - \theta|$$

Claim 2: $x_{(2)} = y_{(2)}$

On contrary, WLOG let $x_{(2)} < y_{(2)}$. On choosing $\theta \in (x_{(2)}, \min\{y_{(2)}, x_{(3)}\})$ and proceeding as in claim we get $x_{(2)} = y_{(2)}$. Thus $x_{(1)} = y_{(1)}$ and $x_{(2)} = y_{(2)}$

Applying induction it can be shown that

$$x_{(i)} = y_{(i)}, \quad i = 1 \ldots n$$

Thus $T(\underline{X}) = (X_{(1)}, \ldots, X_{(n)})$ is minimal sufficient

Using the example considered in (a), it follows that $T(\underline{X})$ is not complete

Problem No. 13 (a) $P_f(N) = 0, \ \forall\ f \in \mathcal{P}_0 \Rightarrow P_f(N) = 0 \ \ \forall f \in \mathcal{P}_1$

$$\mathcal{P}_0 \subseteq \mathcal{P}_1$$

T is sufficient and complete for $\mathcal{P}_0$

Claim: T is complete for $\mathcal{P}_1$.

Let $\psi(\cdot)$ be a function such that

$$E_f(\psi(T)) = 0, \quad \forall\ f \in \mathcal{P}_1$$

$$\Rightarrow \quad E_f(\psi(T)) = 0, \quad \forall\ f \in \mathcal{P}_0$$

$$\Rightarrow \quad P_f(\psi(T) = 0) = 1, \quad \forall\ f \in \mathcal{P}_0$$

$$\Rightarrow \quad P_f(\psi(T) \neq 0) = 0, \quad \forall\ f \in \mathcal{P}_1$$

$$\Rightarrow \quad P_f(\psi(T) \neq 0) = 0, \quad \forall\ f \in \mathcal{P}_1$$

$$\Rightarrow \quad P_f(\psi(T) = 0) = 1,$$

$\Rightarrow$ T is complete for $\mathcal{P}_1$

(b) Let $\mathcal{P}_0 = \{ f_{\underline{\theta}} : \underline{\theta} \in \Theta = \mathbb{R}^n \}$

Here

$$f_{\underline{\theta}}(\underline{x}) = c(\underline{\theta})\, e^{\sum_{i=1}^{n} \theta_i \left(\sum_{j=1}^{n} x_j^i \right) - \sum_{i=1}^{n} x_i^{2n}}$$

$\eta = (\theta_1, \dots, \theta_k) \in \mathbb{R}^n$ contains a n-dimensional open rectangle

$\Rightarrow U(\underline{X}) = \left(\sum_{j=1}^{n} x_j, \sum_{j=1}^{n} x_j^2, \dots, \sum_{j=1}^{n} x_j^n \right)$ is complete-sufficient for $\mathcal{P}_0$

Since roots of equation

$$(\theta - x_{(1)})(\theta - x_{(2)}) \cdots (\theta - x_{(n)}) = 0$$

or

$$\theta^n - \left(\sum_{i=1}^{n} x_{(i)} \right) \theta + \left(\sum\sum_{i \neq j} x_{(i)} x_{(j)} \right) \theta^{n-1} + \dots + (-1)^n x_1 x_2 \cdots x_n = 0$$

are known

$U(\underline{X})$ is a 1-1 fn of $(X_{(1)}, \dots, X_{(n)})$

$\Rightarrow T(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$ is complete for $\mathcal{P}_0$

Also if for some $f_{\theta_0} \in \mathcal{P}_0$

$P_{f_{\theta_0}}(N) = 0$

$\Rightarrow \int_N f_{\theta_0}(\underline{x})\, d\underline{x} = 0 \Rightarrow \mu(N) = 0$, where μ is the Lebesgue measure

$\Rightarrow P_f(N) = 0 \quad \forall f \in \mathcal{P}$

Now using (a) it follows that $T(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$ is complete.